ICAB July-August 2022 Exam
Case Study
Model Answer:

Draft Report

To : Board of Directors, Zenith Agro Business Limited

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Subject : Report on Assessment of business operation, evaluation of strategic options, suggestion of investing and financing options, assisting on IPO, advising on accounting & tax issues, regulatory and secretarial matters.

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1. Executive Summary:

1.1. Evaluation of financial and non-financial performances Zenith Agro.

During the year Zenith has shown an outstanding performance. Its revenue has increased tremendously by 17.2% (BDT 34,571,199) from previous year due to increased market demand. Cost of revenue has increased by BDT 30,476,462 (17.9%) from previous year matching the increase in revenue.

Zenith has enjoyed massive growth of 13.5% (increased by 34,321,721) in its current year gross profit due to significant increase in growth in revenue from previous year. Despite the impressive growth in GP margin, Zenith has performed poorly in terms of net profit. Net profit margin is only 2.8% (BDT 6,592,211) after considering decreasing adjustment in operating expenses.

Zenith has experienced increase in current ratio of 3.3 in 2022 from 1.9 from previous year as the company has significantly invested in its working capital. Zenith has Asset turnover ratio of 1.69 in the year 2022 compared to 1.50 in 2021. However, there are still opportunity to improve efficiency in asset utilization. Zenith has been achieving organic growth by expanding business operations and creating demand for its products.

Company initially reported operating profit of BDT 6,320,605 without properly considering the year end valuation of biological assets, government grants and employee benefit expenses. After adjustment for gain from biological assets of BDT 2,313,760, Government grant of BDT 1,000,000; gain from interest free loan of BDT 555,175 and employee benefit expenses of 1,260,000, operating profit arrives at BDT 8,929,540. For the purpose of simplicity WPPF expense has not been adjusted.

Analysis of the financial statement and ratio analysis shows strong financial condition of the company which it achieved organically. Company has strong current and quick ratio. TIE ratio is very high as the company is not geared. Although asset utilization ratio has increased slightly, Zenith has room for improving efficient utilization of assets to generate revenue.

As Fish production has higher gross profit margin, Zenith should give emphasis on increasing fish production. Zenith should introduce gearing in the company as the gearing ratio is very low. Net profit of the company is too low. Management should decrease operating expenses to increase net profit. Zenith Asset turnover ratio is only 1.5. It should try to increase its efficiency in asset utilization by generating higher revenue. Management should properly incorporate all the necessary adjustment entry at the year end.

Recommendation to the board includes:

- Take initiative to maintain the revenue growth.
- As Fish production has higher gross profit margin, Zenith should give emphasis on increasing fish production.
- Zenith should introduce gearing in the company as the gearing ratio is very low.
- Increase efficiency in generating revenue utilizing Zeniths total assets.

1.2. SWOT Analysis.

Zenith has strong financial position which it can utilize to generate more revenue by increasing efficiency. It has diversified its business operation by introducing fish production and dairy production to the farm which will help Zenith to tackle business challenges. It won award and government awards which created credibility to Zenith leading increase in product demand. After creating strong market presence, Zenith now has stepped in internal market by exporting its products.

One of the key challenges for Zenith is souring fish fry from external parties which may hamper quality of fish production. Its dairy operation requires huge investments for acquiring biological assets, fees, infrastructure, and hygiene. Managing huge amount of cattle wastage produced every day is big challenges as it pollutes environment if not managed.

Despite some weakness in the business production, Zenith has opportunity to related and unrelated diversification. It can open sweet shops and meet shops which can take input from Zenith's own dairy production. Department of Livestock Services has taken various initiative to start exporting meats. In addition, Zenith has plans to start its own 'ready to drink' milk brand.

Like any other business, there are threats to the business. One of the key threats is increased competition from large number of small businesses who producing similar produce. Cost of production may increase if the fuel cost price increases. Workforce who are directly related in production are leaving Zenith in three years with decreasing number of resources coming in agriculture profession.

Conclusion:

- Zenith has strong and proven business model which helps Zenith to achieve organic growth.
- Agricultural produce is highly dependent on input. As input, Zenith produces its own seeds whereas it purchases fish fries from outside.
- Zenith's operating is inherent challenge that its produces are quickly perishable. It needs to be should quickly right after harvesting.

Recommendation:

- Zenith should go for related and unrelated diversifications to grow organically and manage business risks.
- Zenith should start focusing on producing its own fish fries to have more control over quality of fish productions.
- Zenith should focus on improve packaging of to delay perishability of agricultural produces.

1.3. Identification and evaluation on strategic options:

Zenith board is currently considering between two strategic options of setting up a biogas plant and expanding fish production. Board prefers NPV approach to evaluate strategic options and determines cost of capital using CAPM model which results a discount rate of 13%. Biogas option has been evaluated considering project life of 10 years whereas fish production project has been evaluated considering 15 years. Project cash flows has been determined by deducting all annual expenses from annual revenue. It has been assumed that cash flows arise at the end of the year.

After calculation of Net Present Value of project cash follows using 13% discount rate, it has been identified that Biogas plant results in positive NPV of BDT 7,303,965 whereas for fish production options generate negative NPV of 28,913,833.

Recommendation to the board:

- Zenith should opt for implementing biogas project. It will generate additional revenue for the company as well as solve the challenges of managing cattle waste.
- Zenith may redesign the business model for fish production. As fish production provides higher gross profit margin, there may be opportunities for increasing the cash flows which might lead to positive NPV. Furthermore, Zenith may produce fishes which has high demand and sells at fair price.

1.4. Justification for selecting biogas plant for execution:

Both of the projects are in strategic fit for Zenith Agro. Both the project will help Zenith to grow, increase profitability and add shareholder wealth. Zenith is currently not is a position where it can select both of the project to implement. Calculation of NPV shows that setting up biogas plant generates positive net present value. Therefore, it should be selected for implementation.

Recommendation:

- As the NPV of biogas plant option is positive and higher than fish production NPV, Zenith should opt for implementing biogas project as it will create new revenue stream.
- Zenith may take cattle waste from other dairy farms in exchange of money to increase revenue from biogas plant.

1.5. Best option for financing selected project:

Zenith has two options, bank loan and share issue, to finance its project for setting up biogas plant. In either option, Zenith needs to include bank representative or independent director in its board who are required to be paid board meeting attendance fee. In both option, Zenith is required hold quarterly board meetings where quarterly financial statements are required to be presented. This is additional compliance requirement for Zenith. Calculation of project cash flows after adjusting with finance related expenses are discounted with cost of capital. Cost of loan finance has been determined at 12% which provide Net Present value of 11,281,143 of the project cash flows. On the other hand, cost of capital for equity finance has been determined at 13% which provides net present value of BDT 6,761,340. Calculation shows that setting up biogas plant with bank loan will generate higher NPV compared to financing though share capital.

Recommendation to the board:

- NPV from bank financing is BDT 11,281,143 where has NPV from Share capital financing is BDT 6,761,340. As NPV from bank financing is higher, Zenith should finance the biogas setup project though bank loan.
- Zenith should try to increase the net present value by repaying loan on time which will remove the delinquent charge from loan interest. Furthermore, obtaining NOC from existing creditors will remove 2% risk premium what would decrease bank interest rate and increase NPV.

1.6. Assessment four pillars of sustainability

Zenith is focusing on all four areas, human, social, economic and environmental, of sustainability. As part of human sustainability, Zenith ensures that its employees wear safety gears when working at the fields and the dairy house. Employees were always given training on regular basis. Zenith produces vegetables with limited quantity of pesticides considering its negative impacts in human body. As part of social responsibility, Zenith has undertaken initiative to produce pesticide and GMO-free green vegetables. It also has focused on reducing wastage and packaging from recycled materials. It is also creating jobs for the society. To ensure economic sustainability, Zenith make money by supplying foods to consumers. By being profitable, Zenith ensured that it could continue its business. It also creating many job opportunities and contributing to national GDP. As part of Environmental sustainability, Zenith is producing its own seeds and fish fries to ensure continuity of production. It also encourages recycling wastage produces and avoids using plastic material. Despite all these initiatives, Zenith is still disposing the cattle wastage in open land which causing air and water pollution.

1.7. Recommendation to the board

- Zenith should address managing cattle waste with top priority as it is causing environmental pollutions and discomfort to neighbouring families.
- Management should pay overtime to its employees as per labour law.
- Zenith should keep maintaining its intention of providing fresh vegetables without harmful fertilizers and GMOs.

1.8. Assessment of ethical issues.

Zenith appointed a new auditor when prior year auditors have disagreed to provide clear report despite the new auditor has independence threat. It enticed super shop employees to get vegetable supply contract which it is violating by supply third party produced vegetables. Zenith uses chemical fertilizers and pesticides to increase and secure production which cause health issues, deteriorates soil conditions and kill native species. In addition, zenith has unethically denied overtime and employee benefits to its employees despite required by Labor Law, 2006.

Cattle in zenith's dairy are kept in unsanitary conditions which damaging their body and physical condition. Zenith artificially impregnates cows and artificially fed calves depriving them from suckling from the mother cows by milked 100% for human consumption. It disposes the cattle wastage in open land which causing air and water pollution. It is causing difficulties for nearby people to live in with bad smell.

1.9. Recommendation

- Zenith should appoint an independent auditor following all the legal requirements.
- Zenith should immediately take actions to manage its cattle wastes.
- Management should pay overtime and employee benefits to its employees as per labour law.
- It should improve cattle rearing facilities to raise cattle in sanitary and healthy conditions

2. Adjustment for reporting and compliance issues and evaluation of financial and non-financial performances of Zenith. (Requirement-a)

2.1. Adjustments required in the financial statements for reporting and compliances issues

1. Initial recognition of biological assets (calves):

When 20 calves were born at the firm, Zenith should have recognized the calves as it could be reliably estimating the fair value of the calves. At initial recognition (during birth), calves had fair value of BDT 300,000 (20 calves @ 15,000). Management estimates cost to sell will be 1% of fair value which is 3,000 (1% of 300,000). Therefore, fair value less cost to sell is 297,000 which should have been recognized at the time of initial recognition by posting following entry:

Biological Assets	Dr.	297,000	
Gain/loss from biological assets	Cr.		297,000

2. Year-end recognition of biological assets (calves):

At the year end, these calves should be revalued at the fair value at the year end. These calves are valued 25,000 each at the year end. Therefore, calves had fair value of BDT 500,000 (20 calves @ 25,000) at the year end. Estimated cost to sell will be 1% of fair value which is 5,000 (1% of 500,000). Therefore, fair value less cost to sell is 495,000 which should have been reported at the year end. Biological asset value for these calves should increase by BDT 198,000 (495,000 – 297,000) by posting following entry:

Biological Assets	Dr.	198,000	
Gain/loss from biological assets	Cr.		198,000

3. Year-end recognition of biological assets (other than calves):

Biological assets other than the calves is valued at BDT 22,479,045. After 1% expected selling cost, fair value less cost to sell at the year-end stands at BDT 22,254,255. These biological assets have been reported at BDT 20,435,495 at the year-end financial statements. Hence, Biological asset has to be adjusted by BDT 1,818,760 (22,254,255- 20,435,495) and following entry should be posted:

Biological Assets	Dr.	1,818,760	
Gain/loss from biological assets	Cr.		1,818,760

4. Government Grant:

Zenith received cash reward of BDT 1,000,000 from government for its contribution in Fish Farming. As the condition for grant has already been fulfilled, Zenith can immediately recognize the grant. Zenith should post following entry for award receivable.

Accounts receivables	Dr.	1,000,000	
Other income	Cr.		1,000,000

Zenith is also expecting to receive 20% cash incentives on its export proceeds. However, receipt of such incentive is conditional on due diligence clearance by the authorized dealer and the central bank. As these conditions are not yet fulfilled and these cash incentive receivables are contingent assets until approved by Authorized dealer and the central bank, Zenith should not recognize the cash incentive amount.

5. Interest Free Government Loan:

Zenith received interest free loan of BDT 2,500,000 which is repayable at 500,000 per year. In accordance with IFRS 9, this should be recognized at amortized cost. In absence of implicit interest rate, incremental borrowing rate may be considered. Zenith is planning to take bank loan which may be used to discount the loan. Considering that loan will be paid at the end of the year at 9% interest rate, present value of the loan is 1,944,825 which should be reported as loan liabilities. Differential amount of 555,175 will be considered as government grant. At the end of the year, interest will be charged on the outstanding loan amount. As per IAS 1, the loan liabilities will be classified between current portion at BDT 458,716 and non-current portion of BDT 1,486,110. Following adjustment entry should be posted to adjust the reported loan as short-term borrowings.

Short Term Borrowings	Dr.	1,866,250	
Interest Expenses	Dr.	175,035	
Long-Term Borrowing	Cr.		1,486,110
Other Income	Cr.		555,175

6. Employee Benefits:

Zenith has 60 permanent workers who are not given overtime or post-employment benefits. These workers work 2 hours in excess of regular hours for 5 days a month. Which means per employee works extra hours for 5 x 12 = 60 days a year. Hourly overtime rate is BDT 50 (BDT 200 per day x 2 /8 hour). Total overtime expense should be paid to the works is (50 taka per hour x 2-hour overtime x 60 days x 60 workers) = 360,000.

Zenith is also required to pay gratuity expenses as per the labour law, 2006. As per the labour law, gratuity is payable to a permanent worker who worked for 6 month or more in a year. For each year worked, workers will be eligible for gratuity equivalent of monthly compensation. Zenith pays monthly compensation of (200 taka per day x 25 days) = BDT 5,000 per worker and each worker works for 3 years on average. Hence total gratuity liability for 60 workers will be (5,000 x 3 x 60) = BDT 900,000.

To recognize the overtime expense and gratuity liabilities, Zenith should post following entry:

Overtime Expenses	Dr	360,000	
Gratuity Expenses	Dr	900.000	
Overtime Payable	Cr.		360,000
Gratuity Payable	Cr		900,000

2.2. Evaluation of financial after adjustment

Revenue:

Zenith's revenue has increased tremendously by 17.2% (BDT 34,571,199) from previous year revenue of BDT 200,804,966 leading to total Revenue of 2022 at 235,376,165. This revenue growth is mostly contributed by the dairy operations which increased by 19,014,159 (17.2%). Other revenue streams like vegetable production and fish production also contributed to the revenue growth. In revenue mix, dairy operation contributes 55%, Vegetable production contribution 30% and Fish production contributed 15% of total revenue. During the year Revenue from Fish production has increased by 25.6% which shows significant growth potentials and could be leading contributor to revenue in future. Vegetable production and dairy operation achieved growth in 2022 by 13.4% and 17.2% respectively which are also impressive.

Cost of Sales:

Cost of Sales in 2022 was BDT 201,054,444. It has increased by BDT 30,476,462 (17.9%) from previous year cost of revenue of 170,577,982. Cost of sale has increased at higher rate compared to revenue growth rate. This is due to increase in direct costs like labor transportation costs associated with vegetable production and dairy operation. Of the total cost of sales, 55% (BDT 110,579,944) is incurred by Dairy production, 31% (BDT 62,326,878) by vegetable production and 14% (BDT 28,147,622) by fish production. This is in line with the revenue contribution percentage. Growth in cost of sales also follows revenue growth rate.

GP and GP Margin:

As revenue has experienced significant increased, Zenith witnessed handsome amount of growth 13.5% (increased by 34,321,721) in its gross profit. In 2022, gross profit of Zenith was BDT 30,226,984 which was BDT 34,321,721 in the year 2021. Growth of gross profit is arising from significant increase in Revenue. However, GP margin has decreased to 14.58% compared to 15.05% from previous year.

Gross Profit in 2022 was mostly contributed by Dairy production (55%). Vegetable production and fish production has contributed 24.1% and Fish production has contributed 21% in gross profit. This is brilliant achievement for Zenith. GP growth in 2022 has been supported by19.7% growth in dairy products. Vegetable production and Fish production has contributed only 8.1% and 5.4% in GP growth margin. GP for Zenith has increased despite Zenith experienced slightly higher cost of sales growth rate. Overall, Zenith was successful in making handsome amount of gross profit.

Net Profit Margin

Despite brilliant sales growth and gross profit margin, Net profit of Zenith is very poor. During the year, adjusted net profit margin is 2.8% and net profit was BDT 6,592,211 whereas net profit margin in 2021 was 1.08% (2,165,003). Despite significantly low net profit margin, net profit for Zenith has increased by 204% to BDT 6,592,211 from previous BDT 2,165,003. This increase in profitability is due to decrease in Finance Charge and Deferred Tax income. It is assumed that adjustment for accounting issues have not impacted WPPF and tax expenses.

Current and Quick Ratio

Zenith has experienced increase in current ratio of 3.3 in 2022 from 1.9 from previous year. This shows that Zenith has significantly invested in its working capital, and it has enough current asset to meet its current obligation. Increase in working capital was caused by payment of short-term borrowings.

Quick ratio of Zenith also increased to 1.8 from 0.8 of previous year. This indicates that Zenith has sufficient amount of liquid asset to pay off the current liabilities if they fall due. However, working capital seems higher and may be sit idle in the company.

Debt to Equity and TIE Ratio

Gearing ratio for Zenith has not charged significantly as Zenith has doesn't have significant amount of debt. It has debt to equity ratio of 0.24 in 2022 and 0.23 in year 2021. Finance expense has decreased from previous year which has contributed to increase in TIE ratio.

TIE ratio has increased to 11.6 in 2022 compared to 3.2 in 2021. In 2022 Zenith was able to generate operation profit of 11.6 time of the finance expenses whereas in 2020 net profit was only 3.2 times of the finance expenses. With the sufficient profitability, Zenith have strong position to face challenge of debt serving. Zenith is highly ungeared company. It may increase gearing to increase profitability.

Asset Turnover and Return on Assets

Zenith has Asset turnover ratio of 1.69 in the year 2022 compared to 1.50 in 2021. Although there has been slight increase in Asset turnover ratio, but the ratio is still very poor. Zenith was able to generate revenue nearly 1.5 times of its Assets which it uses to generate revenue and profit. This means Zenith efficient enough to generate revenue utilizing its assets. However, there are still opportunity to improve efficiency in asset utilization.

Return on asset is (4.7%) in 2022 compared to 1.6% of 2021. Return on asset has increased significantly because of decrease in finance expense, tax expense and increase in operating profit. Because of these positive changes, profit for the company improved significantly which led to increase in return on asset. If Zenith could improve efficiency in asset turnover, return on asset could have increase significantly.

Return on Equity

In 2022, return on equity has increased to (6.6%) in 2022 compared to 2.3% in 2021. This massive increase is due to significantly increase in net profit for the year. This rate of return is very low compared to government risk free rate.

2.3. Evaluation of non-financial performance

Although Zenith started business with production of vegetable only, it has grown its business significantly over the time by expanding business to dairy production and fish production. In its original vegetable farming operation, it has increased its operation significantly by introducing new and luxury vegetables. It has also increased its market coverage from local market to Karwan bazar, Dhaka.

It has also made deals with super shops to deliver vegetables, dairy products and fish. This has created significant demand for Zenith products and increased sales as well as profitability.

For its outstanding contribution in fish production, Zenith has received award and government grant. This is outstanding performance by Zenith which helps to create awareness of Zenith's operation as well as product demand.

2.4. Conclusion

- Revenue has increased tremendously by 17.2% (BDT 34,571,199) from previous year due to increased market demand.
- Cost of revenue has increased by BDT 30,476,462 (17.9%) from previous year matching the increase in revenue.
- Zenith witnessed handsome amount of growth 13.5% (increased by 34,321,721) in its gross profit due to significant increase in growth in Revenue from previous year.
- Net profit of Zenith is very poor. During the year, adjusted net profit margin is 2.8% and net profit was BDT 6,592,211 despite considering decreasing adjustment in operating expenses.

- Zenith has experienced increase in current ratio of 3.3 in 2022 from 1.9 from previous year as the company has significantly invested in its working capital.
- Zenith has Asset turnover ratio of 1.69 in the year 2022 compared to 1.50 in 2021. However, there are still opportunity to improve efficiency in asset utilization.
- Zenith has been achieving organic growth by expanding business operations and creating demand for its products.

2.5. Recommendation

- Zenith has been enjoying outstanding performance. It should focus on maintaining revenue growth.
- As Fish production has higher gross profit margin, Zenith should give emphasis on increasing fish production.
- Zenith should introduce gearing in the company as the gearing ratio is very low.
- Net profit of the company is too low. Management should decrease operating expenses to increase net profit.
- Zenith Asset turnover ratio is only 1.5. It should try to increase its efficiency in asset utilization by generating higher revenue.
- Management should properly incorporate all the necessary adjustment entry at the year end.

3. Evaluate of the Strengths, Weaknesses, Opportunities, and Threats (SWOT) of Zenith's existing business (Requirement B-i):

3.1. SWOT Analysis:

Strengths

- Zenith is an award-winning company for fish production. It creates credibility to Zenith which will lead increase in product demand.
- Zenith has strong financial position which it can utilize to generate more revenue by increasing efficiency.
- Zenith has diversified its business operation by introducing fish production and dairy production to the farm. This will help Zenith to tackle any forthcoming challenges in market demand.
- Zenith has strong organic growth rate. It has achieved revenue growth in 17.2% in 2022. It has been consistently achieving huge revenue growth over the years.
- Zenith has strong market presence. It has grown from local market to Karwan bazar and Super shops. It also started exporting in international market.
- Zenith is self-dependent regarding sourcing of seeds for vegetables.
- Zenith uses various technology in its operation to boost up its productivity and increase efficiency.
- Zenith has easy access to raising finance.

Weakness

- Zenith often sources vegetables from local market to fulfil obligation to super shops. However, it doesn't have control over quality and production of those vegetables.
- Quality of fish production is highly depended on the fish fry. As Zenith most of the demand for fish fry is met by sourcing from external source, production may be hampered if quality of purchased fries are not up to the standard.
- Dairy farming is capital intensive as the cattle itself require huge investment on biological assets, fees, infrastructure and hygiene.
- Agro produces are quickly perishable. It may not be in sellable condition if it could not be sold timely manner.
- Managing huge amount of cattle wastage produced every day is big challenges as it pollutes environment if not managed.

Opportunities

- Zenith has further opportunities for related and unrelated diversifications in future.
- Zenith has opportunity to open sweet shops and meet shops which can take input from Zenith's own dairy production.
- Zenith plans to start its own milk brand where it will sell pasteurized and 'ready to drink' milks under the brand 'Zenith Dairy Milk.
- Zenith has opportunities to increase its export of agro produces including fish and meat.
- Opportunity to expand cattle sale operation focusing on Qurbani eid.
- Opportunity to raise finance through IPO, QIO and Bank loan.
- Zenith can start selling seeds to other agro businesses.
- Zenith has opportunity to produce biogas which can be used for road vehicles. As fuel price is expected to increase, there will be demand for alternative energy sources. Biogas produced by zenith may be a source of alternative energy.

Threats

- Cost of fuel may increase in next couple of months which will increase transportation costs and fuel expense and leading to decrease in profitability.
- Survival of fries when released in the ponds. Significant fluctuation in pH level in the pond water can be damaging in fish production.
- Many smaller farms have popped out throughout the country which would increase market competition.
- Employee turnover ratio is high. Most of the workers work on the field works for only three years on average. This increases recruitment and training cost of hiring training the employees.
- If Zenith cannot manage waste from cattle farm, it will cause environmental damage and may attract financial and non-financial penalty.

3.2. Conclusion:

- Zenith has strong and proven business model which helps Zenith to achieve organic growth.
- Zenith has diversified its business operation by introducing fish production and dairy production to the farm.
- Agricultural produce is highly dependent on input. As input, Zenith produces its own seeds whereas it purchases fish fries from outside.
- Zenith's operating is inherent challenge that its produces are quickly perishable. It needs to be should quickly right after harvesting.

3.3. Recommendation:

- Zenith should go for related and unrelated diversifications to grow organically and manage business risks.
- Zenith should start focusing on producing its own fish fries to have more control over quality of fish productions.
- Zenith should put focus on managing cattle waste.
- Zenith should focus on improve packaging of to delay perishability of agricultural produces.

4. Analysis of the strategic options available to the board of directors (Requirement B-ii):

Zenith board of directors are currently considering between two investment options. First option is setting up a biogas plant which will take cattle waste as input in producing biogas. Second option is to expand its fish production which generates highest gross profit margin. Biogas plant will solve the challenge Zenith is facing with cattle waste. On the other hand, fish production will increase profitability for Zenith.

Board apprises investment options using a discount rate determined by CAPM model. Board has identified that risk free rate is 7% where market return is 12% from similar companies. Also, board identified that marker risk beta is 1.2. Using these information, CAPM model provides a discount rate of 13% [*e.g. 7% + 1.2(12%-7%)*] which shall be used to apprise the investment options. For the purpose of calculation, it has been assumed that cash flows arise at the end of the year except for investing outflows which occurs at the beginning of the year.

4.1. Analysis of strategic option to set up biogas plant.

In this option, Zenith will require to invest 70,000,000 for biogas plant machineries and further 30,000,000 for civil works. Once set up is complete, biogas plant will produce 1100 cubic meter of gas per day which can be used to fill 110 vehicles or local households. Per liter gas will be sold at 50 taka. Therefore, annual revenue of gas will be 365 x 1100 x 50 = 2,00,75,000.

To maximize the useful life of the machineries to 10-year, regular serving will be required which will incur cost of BDT 25,000 per month or BDT 3,00,000 per year. Therefore, net cash flows per year till the end of useful life will be 19,775,000. Useful life of the machines are 10 years which has been used to apprise the project. It has been assumed that there will be no terminal value at the end of 10 year.

As calculated in ***Annexure 4***, net present value of setting up biogas plant is 7,303,965. As the net present value discounted at 13% provide positive cash flows, that means options to set up biogas plant is acceptable.

4.2. Analysis of strategic option to expand fish production operation.

Second option is to expand current fish production as this generates highest revenue and gross profit. Under this project, Zenith will require BDT 80,000,000 to build infrastructure for hatchery and feed production and another BDT 20,000,000 for fish operation. Total cost at initial period will be BDT 100,000,000.

It will incur cost of 9,000,000 to generate revenue of BDT 20,000. Developed infrastructure will have useful life of 15 years which has been considered as project life to apprise the options. It has been assumed that there will be no terminal value at the end of 15 year.

As calculated in Annexure 5, net present value of the fish project yields negative NPV of BDT 28,913,833. As the NPV is negative, therefore this project should not be accepted.

4.3. Conclusion:

- Zenith uses CAPM model to determine the discount rate for apprising strategic options. Calculation of cost of capital using CAPM model provides 13% discount rate.
- Biogas option has been evaluated considering 10 years whereas fish production project has been evaluated considering 15 years considering the useful life of machines and infrastructure. It has been assumed that there will be no terminal value at the end of useful life of the machines and infrastructure.
- Cash flows has been determined by deducting all annual expenses from annual revenue. It has been assumed that cash flows arise at the end of the year.
- Net present value of Biogas plant is BDT 7,303,965 whereas for fish production options generate negative NPV of 28,913,833.

4.4. Recommendation:

- As the NPV of biogas plant option is positive and higher than fish production NPV, Zenith should opt for implementing biogas project. It will generate additional revenue for the company as well as solve the challenges of managing cattle waste.
- Zenith may redesign the business model for fish production. As fish production provides higher gross profit margin, there may be opportunities for increasing the cash flows which might lead to positive NPV. Furthermore, Zenith may produce fishes which has high demand and sells at fair price.

- Zenith should consider impact of all variables and assumptions while determining cash flows and apprising strategic options.
- Zenith should finance biogas plan option by using a source which has lower cost of capital.

5. Justification for option implantation (Requirement B-iii):

Zenith has around 80 cattle in its dairy operation. Which produces huge amount of cattle waste every day. Zenith has been facing challenges to manage these cattle produces. They were not able to sell dried waste to local households as fuel. Currently it has been dumping the cattle waste in open field which is polluting the environment. Zenith already received warning from local council to address the issue as soon as possible. However, biogas project is entirely new to Zenith. Although it is expected to resolve the crisis Zenith facing, management has no experience in producing biogas and conducting biogas operation. Management may have to hire experienced employees to run the biogas project.

On the other hand, fish production option was generating highest gross profit margin despite being a comparatively new operation within Zenith. It has already received an award and government grant for its contribution in fish production. It is only logical for management to expand this highly profitable and well recognize segment of their business. Management is already experienced with fish production. So, scaling up will not lead to a challenge for the management.

It shows that both of the projects are in strategic fit for Zenith Agro. Both the project will help Zenith to grow, increase profitability and add shareholder wealth. However, Zenith is currently not is a position where it can select both of the project to implement. Zenith board has decided to apprise both of the strategic options by calculating NPV. Strategic option that will yield more net present value would be selected for implementation. From the calculation of NPV in earlier, it was noted that setting up biogas plant generates positive net present value whereas fish expansion project indicates negative net present value. Therefore, it is only logical for Zenith management to select biogas plant project for implementation and set aside fish expansion project for now.

5.1. Conclusion:

- Both biogas plant project and fish production expansion project fits with Zenith's strategic position. Biogas plant will help Zenith to address its concern related to management of cattle waste. Whereas fish production project will help Zenith to increase profitability for the company.
- Board apprises projects using NPV method after discounting at rate determined by CAPM model.
- NPV from biogas plant are higher comparative to the fish production project.

5.2. Recommendation:

- As the NPV of biogas plant option is positive and higher than fish production NPV, Zenith should opt for implementing biogas project.
- Biogas plant will create a new revenue stream for the company while addressing the challenges of managing cattle waste.
- Zenith may take cattle waste from other dairy farms in exchange of money to increase revenue from biogas plant.

6. Advice on financing options (Requirement b-iv).

Zenith board can manage finance of BDT 100,000,000 to invest in the strategic options. Board currently has two options to raise the require fund. First option is to raise the fund to bank loan and second option is to raise fund through QIO.

In case of bank loans, bank has few terms and conditions in addition to delinquent charge, commitment fee and risk premium in absence of NOC. One of the conditions is to appoint two independent directors in company board who are required to pay fees of BDT 10,000 per meeting per person for each of the quarterly board meetings. This is an additional expense which need to be adjusted from project cash flows. Furthermore, Zenith need to provide personal guaranty of the shareholders. This is an additional requirement for the loan, and this might make the borrowing process complicated. However, Zenith needs to comply to get the loan. Zenith is planning to take the loan for 10 years.

6.1. Financing through bank:

Management has communicated with its bank which has offered loan at an interest rate of 8.5% per annum. Bank will further charge 1% delinquent charge over the interest rate for overdue instalments. However, management believes it should consider the delinquent charge as it has liquidity limitations. Bank also will charge 0.5% as commitment fee for unutilized loan. As management believes full amount of loan will be withdrawn at the beginning of the year, there will not be any unutilized funds.

Bank also requested a 'No Objection Certificate (NOC)' from the existing lenders to create priority charge on company assets over existing creditors. In absence of NOC to the bank, additional 2% 'risk premium' will be added on outstanding loan amount. Zenith management believes it would not be able to provide an NOC to the bank and hence will be required to pay 2% additional interest over the loan amount. Therefore, appropriate bank borrowing rate will be 8.5% + 1% + 2%= 11.5%. We may round off the interest rate to 12% for simplicity in calculation.

To comply with the bank requirement, Zenith will have to appoint two independent directors who will participate in quarterly meeting and will be paid BDT 10,000 per meeting. That means Zenith will have to pay (10,000 x 2 x 4) BDT 80,000 taka per year as board meeting attendance fee to these independent directors. These outflows should be adjusted with the project cash flow as these are relevant to project financing. As previously calculated, annual cash flow from biogas was BDT 19,775,000 from which BDT 80,000 should be adjusted. Therefore, revised cash flow should be 19,695,000 which would be discounted at 12% interest rate.

As shown in Annexure 6, net present value of biogas project after financing through bank loan is 11,281,143. If the NPV of other financing options is below 11,281,143, financing though bank loan will be appropriate.

6.2. Financing through QIO:

Board is currently considering raising fund through QIO where it plans to issue shares at 60% premium i.e., 6 taka over face value. Total issue price will be 16 taka per share subject to approval of BSEC which considers NPV, EPS and Sectoral Market PE ratio.

Following the NPV method, share price would be (Adjusted Total Asset/ Total shares outstanding = 139,347,127 / 10,000,000) = 13.93 per share. Latest EPS reported was 0.42. However, after posting the adjustment, revised EPS is 0.66. Company with similar EPS has share price of BDT 15.30. On February 2022, Sectoral PE Ratio was 23.18. Based on sectoral PE ratio and revised EPS of Zenith, indicative share price for Zenith is (23.18 x 0.66) = 15.30. Average share price of similar companies is (35+28_12+18+7)/5 = 20 taka. Considering share price based on NPV, EPS, Sectoral PE and average share price, it is highly likely that BSEC will approve 60% share premium. Therefore, it is likely that Zenith will be able to issue share at a price of 16 taka.

When listed in DSE SME Platform, Zenith need to appoint independent director at the ratio of $1/5^{th}$ of its current board members. That means for every 4 dependent director there should be 1 independent director. As Zenith has already 5 directors in its board, it needs to appoint 2 independent directors in its board who are required to pay 10,000 taka each for every quarterly board meeting and AGM is attended by them. In a year there would be 4 quarterly board meeting and 1 AGM. So, meeting attended fee payable to the independent directors are (5 x 10000 x 2) =100,000 per year which needs t0 be adjusted from annual forecasted cash flow of 19,775,000. These cash flows than need to be discounted with cost of equity shares. If fund raised though Shares, cost of equity share based on CAPM mode will be, 13% [*e.g., 7% + 1.2(12%-7%)*].

As shown in Annexure 7, net present value of biogas project after financing through equity shares is 6,761,340. NPV of cash flows financed through share is lower in comparison to NPV of cash flows from financing through loan to select best financing option.

6.3. Conclusion

Zenith has two options to finance its project for setting up biogas plant. First option is to finance through bank loan and second option is to finance through share capital. Both the option has its pros and cons. In either option, Zenith needs to appoint independent director and hold board meeting per quarter. In each option, Zenith have to go through lots of compliance options which may be costly to comply with. One of the benefits of bank loan is that Zenith may get out of compliance requirement once the bank loan is repaid. However, in terms of listing with SME platform, Zenith will have to comply with the compliance requirements forever unless it delist itself from the market.
If the project is financed through bank loan, zenith has to pay 12% interest after considering all the interest factors. With cash flows adjusted for board meeting expenses, annual cash flows from bank financed project arrives at BDT 19,695,000 per year for 10 years which are then discounted at 12% to calculate net present value of the cash flows. Net present value of Biogas project financed by bank loan is BDT 11,281,143.

On the other hand, cost of share capital has been calculated at 13% which has been used to discount the adjusted annual cash flow for board meeting expenses. It has been considered that BSEC will approve the issue of shares at 60% premium has indicative share price falls between 13.93 to 15.30 based on NAV, EPS and PE ratio. If financed through QIO, adjusted annual cash flow will be BDT 19,675,000 which has been expected to be constant during project life of 10 years. Net present value of the cash flows financed through share capital has been calculated at BDT 6,761,340 which is lower in comparison to NPV of cash flows from financing through loan to select best financing option.

6.4. Recommendation

- NPV from bank financing is BDT 11,281,143 where has NPV from Share capital financing is BDT 6,761,340. As NPV from bank financing is higher, Zenith should finance the biogas setup project though bank loan.
- As Zenith should plan for Bank loan financing, it should go through the compliance requirement again to make sure no compliance requirement is unchecked and unfulfilled.
- As part of requirement, Zenith is required to appoint independent directors. Zenith should discuss about qualifications and experience of the representatives from bank in their board.
- Bank should try to forecast cash availability to repay loan instalments in due time. This will remove the delinquent charge. This will decrease the discount charge which will increase the net present value of the project cash flows.
- Zenith should discuss with its existing creditors to obtain NOC which would enable of the bank to create charge over existing creditors. This too would also decrease the discount rate and lead higher net present value.

7. Assessment four pillars of sustainability (Requirement C-i).

In order to be sustainable, an entity need to focus on four areas, Human Sustainability, Social Sustainability, Economic Sustainability and Environmental Sustainability. When all four areas become sustainable, entity achieves sustainability.

7.1. Human Sustainability:

Human sustainability aims to maintain and improve the human capital in society by investments in the health and education systems, access to services, nutrition, knowledge and skills. As part of human sustainability, Zenith takes good care of its employees. The company also have focus on employee safety. It ensures that its employees wear safety gears when working at the fields and the dairy house. Employees were always given training on regular basis. It also ensured first aid kits throughout Zenith premises.

Availability of labor with a cheap wage rate also foster the growth of agro-industries. As most of the rural side labor of this country has farming experience, so they can work easily with a little training. Zeniths has employed more than 80 people to maintain a healthy number of human resources. As a smaller number of people are coming in agro business, Zenith should take steps to ensure availability of human resources for future operations.

Zenith produces vegetables with limited quantity of pesticides considering its negative impacts in human body. Furthermore, Zenith regularly tries to produce variety of foods considering the nutritional value of the vegetables.

Similarly, Zenith avoids injecting steroids and hormones to its cattle that are harmful for human consumption. Zenith is also training village women about rearing cattle. These initiatives show Zenith's dedication for human sustainability.

7.2. Social Sustainability:

Social Sustainability is about identifying and managing business impacts, both positive and negative, on people. Social sustainability aims to preserve social capital by investing and creating services that constitute the framework of our society. The concept accommodates a larger view of the world in relation to communities, cultures, and globalisation.

As part of corporate social responsibility, Zenith has undertaken initiative to produce pesticide and GMO-free green vegetables. Zenith understand that it has responsibilities to the society in which it operates its business. It also has focused on reducing wastage and packaging from recycled materials. As part of social contribution, Zenith provided dry manure to local people so address their fuel crisis in cooking.

In addition to direct social contribution, Zenith has indirectly contributed to the society. It is producing fish, vegetable, dairy and meet and supplying to the society as fresh food. Because of its contribution, society is not facing food crisis. Furthermore, Zenith is creating jobs which is also creating social value.

Zenith conducting is business complying with the local and national laws and regulations, this created a compliance environment with strengthen social sustainability. Training rural women in cattle rearing and purchasing milk from them empowers women by becoming self-dependent. This also contribution to society.

7.3. Economic Sustainability:

Economic sustainability aims to maintain the capital intact. If social sustainability focuses on improving social equality, economic sustainability aims to improve the standard of living. In the context of business, it refers to the efficient use of assets to maintain company profitability over time.

Primary objective of Zenith Agro is to supply foods to consumers and make money from it. Since formation, Zenith has expanded by having organic growth. That means it had maintained profitability over time. Trend analysis for last 5 years shows that it has increased its EPS gradually in every year.

By being profitable, Zenith ensured that it could continue its business for foreseeable period and grow its business over time. As Zenith grows and continues operation, it can create many job opportunities which can address unemployment problem in the society. This will also increase economic activities in the country which will eventually contribute to national GDP.

Zenith has also started exporting in foreign countries which will create international market for vegetables, fish, meat and foods produced in Bangladesh. This way Zenith and other Agro businesses will be earn foreign currencies for Bangladesh. This will boost up national economy.

7.4. Environmental Sustainability:

Environmental sustainability aims to improve human welfare through the protection of natural capital (e.g., land, air, water, minerals etc.). Initiatives and programs are defined environmentally sustainable when they ensure that the needs of the population are met without the risk of compromising the needs of future generations.

Zenith is producing vegetables and fishes using natural resources. It is producing its own seeds and fish fries to ensure continuity of production. As the agro produces are quickly perishable, Zenith tries to make these available for consumers for their quick consumption. It also encourages recycling

wastage produces and avoids using plastic material as much as possible to reduce negative impact to the environment. It feeds excess vegetables to cows and wastages as fertilizer in the fields.

Despite all these initiatives, Zenith has not able to stop harming the environment. Zenith disposes the cattle wastage in open land which causing air and water pollution. Cattle farming is one of the largest contributors to greenhouse gases, thus being a major cause of climate change. Cattle waste contains a lot of nitrogen and ammonia, which can contaminate water sources around farms over time. Furthermore, it is causing difficulties for nearby people to live in with bad smell.

7.5. Conclusion

Zenith is focusing on all four areas, human, social, economic and environmental, of sustainability. As part of human sustainability, Zenith ensures that its employees wear safety gears when working at the fields and the dairy house. Employees were always given training on regular basis. It also ensured first aid kits throughout Zenith premises. Zenith produces vegetables with limited quantity of pesticides considering its negative impacts in human body. it avoids injecting steroids and hormones to its cattle that are harmful for human consumption.

Social Sustainability is about identifying and managing business impacts, both positive and negative, on people. As part of social responsibility, Zenith has undertaken initiative to produce pesticide and GMO-free green vegetables. It also has focused on reducing wastage and packaging from recycled materials. It is producing fish, vegetable, dairy and meet and supplying to the society as fresh food. Furthermore, Zenith is creating jobs for the society.

Economic sustainability aims to maintain the capital intact. Zenith supply foods to consumers and make money from it. This adds to the capital of the company and helps zenith in achieving organic growth. By being profitable, Zenith ensured that it could continue its business for foreseeable period and grow its business over time creating many job opportunities. This helps to address unemployment problem of the society and contribute to national GDP.

Environmental sustainability aims to improve human welfare through the protection of natural resources. Zenith is producing vegetables and fishes using natural resources. It is producing its own seeds and fish fries to ensure continuity of production. It also encourages recycling wastage produces and avoids using plastic material. Despite all these initiatives, Zenith has not able to stop harming the environment. Zenith disposes the cattle wastage in open land which causing air and water pollution.

7.6. Recommendation

- Zenith should address managing cattle waste with top priority as it is causing environmental pollutions and discomfort to neighbouring families.
- Zenith should try recycling its wastes in more efficient manner.
- Management should pay overtime to its employees as per labour law.
- Management should avoid using illegal pesticides and insecticides.
- Zenith should keep maintaining its intention of providing fresh vegetables without harmful fertilizers and GMOs.

8. Assessment of ethical issues in Zenith's business operation (Requirement C-ii).

8.1. Ethical issues:

Zenith has appointed KKC as its auditor. Its partner is relative of the directors of the company and helping Zenith in preparation of financial statements and filling tax returns. Auditor has appointed auditor who have independence threat. Furthermore, Prior year auditor has refused to issue and unmodified opinion for which Zenith has changed auditors. New auditor, KKC, has accepted the engagement without taking professional clearance from previous auditors. Moreover, new auditor has not confirmed opening balances which may lead to incorrect reporting by Zenith in its financial statements.

Zenith made deal with super shops to supply own grown fish and vegetables. However, when production is short, Zenith supply vegetables to these super shops after sourcing from external sources. This is complete violation of agreement by Zenith Management. Zenith also often resorts to enticing to super shop officials to buy more or to ensure preference to zenith produces over other agro business or farms.

Zenith has missed several adjustment entries in the financial statements. These adjustments may be intentionally missed out by Zenith management to understate profitability and avoid taxation.

Zenith have 60 permanent workers who works on the vegetable fields, fish and dairy firm. These workers are given daily wages only. They are not paid any overtime or post-employment benefit despite a requirement of Labor Law, 2006. This is an unethical practice by Zenith Management.

Although Zenith prefer uses of organic manure, but still depends on chemical fertilizers and pesticides to increase and secure production which cause health issues. It also deteriorates soil conditions and kill native species, like the honeybees that helps in pollination.

In Zenith dairy, only a small percentage are given access to meadow. Cows are stationed on concrete floors for a long period, damaging their hooves and causing sore joints. Furthermore, these cattle are kept in unsanitary conditions, amidst their own feces, and are forced to inhale toxic fumes. In dairy operation, calves are artificially fed depriving them from suckling from the mother cows and mother cows are milked 100% for human consumption. Moreover, cows are impregnated continuously to ensure they are always lactating. Zenith resorts to artificial insemination to ensure genetically superior and profitable offspring.

Zenith disposes the cattle wastage in open land which causing air and water pollution. It is causing difficulties for nearby people to live in with bad smell. There has been objection filled to local authority against Zenith. Zenith bribed staffs at local council for not taking any action.

8.2. Conclusion

Zenith has portrayed its intention to conduct its business in ethical way. However, its intentions are not supported by its actions. It has changed auditors when prior year auditors have disagreed to provide clear report. It has appointed a new auditor despite the new auditor has independence threat. As a result, the new auditor is not performing its duties in ethical manner. This might lead to intentionally ignoring financial transaction to misstate financial statements to avoid taxation.

Zenith enticed super shop employees to get contract to supply fresh vegetables from own field. However, when production is short, Zenith purchase from third parties to supply to the super shops violating contract agreement. In addition, zenith has unethically denied overtime and employee benefits to its employees despite required by Labor Law, 2006.

Zenith uses chemical fertilizers and pesticides to increase and secure production which cause health issues, deteriorates soil conditions and kill native species. It disposes the cattle wastage in open land which causing air and water pollution. It is causing difficulties for nearby people to live in with bad smell. Zenith bribed staffs at local council for not taking any action.

In Zenith dairy, cattle are kept in unsanitary conditions, amidst their own feces, and are forced to inhale toxic fumes which damaging their hooves and causing sore joints. Furthermore, Zenith artificially impregnates cows to ensure they are always lactating. Calves are artificially fed depriving them from suckling from the mother cows and mother cows are milked 100% for human consumption. This is completely unethical treatment by Zenith.

8.3. Recommendation

- Zenith should change its auditor once more and select an independent auditor following all the legal requirements.
- Zenith should immediately take actions to manage its cattle wastes that causing environmental pollutions.
- Management should pay overtime and employee benefits to its employees as per labour law.
- Zenith should supply own-grown fresh vegetables to super shops as per contract.
- Zenith should be more careful is using chemical fertilizers and pesticides so that it does not or at least reduces its negative impact to health, soil and native species.
- In Zenith dairy, should improve cattle rearing facilities to raise cattle in sanitary and healthy conditions
- Zenith should stop artificially impregnates cows and stop depriving calves from suckling from the mother cows.
- Follow accounting and reporting standards to properly present and report the financial statements.

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9. Appendices:

Appendix 1: Analysis of Adjusted Statement of Profit & Loss

	30-Jun-22	Adjustment	Adjusted 30-06-2022	30-Jun-21	Change	Change %
Sales	235,376,165	-	235,376,165	200,804,966	34,571,199	17.2%
Less: Cost of Sales	(201,054,444)	-	(201,054,444)	(170,577,982)	(30,476,462)	17.9%
Gross Profit	**34,321,721**	**-**	**34,321,721**	**30,226,984**	**4,094,737**	**13.5%**
Less: Operative expenses	(28,001,116)	2,608,935	(25,392,181)	(25,624,253)	232,072	-0.9%
Net operating profit	**6,320,605**	**2,608,935**	**8,929,540**	**4,602,731**	**4,326,809**	**94.0%**
Less: Finance Charges	(592,615)	(175,035)	(767,650)	(1,422,133)	654,483	-46.0%
Profit before WPPF & Tax	**5,727,990**	**2,433,900**	**8,161,890**	**3,180,598**	**4,981,292**	**156.6%**
Less: Provision for WPPF	(286,400)	-	(286,400)	(159,030)	(127,370)	80.1%
Profit before Tax	**5,441,590**	**2,433,900**	**7,875,490**	**3,021,568**	**4,853,922**	**160.6%**
Less: Current Tax Expenses	(1,866,482)	-	(1,866,482)	(1,787,951)	(78,531)	4.4%
Deferred Tax income	583,203	-	583,203	931,386	(348,183)	-37.4%
Profit for the year	**4,158,311**	**2,433,900**	**6,592,211**	**2,165,003**	**4,427,208**	**204.5%**
Other Comprehensive Income	-	-	-	-	-	0.0%
	4,158,311	**2,433,900**	**6,592,211**	**2,165,003**	**4,427,208**	**204.5%**

Appendix 2: Adjusted Balance Sheet

	30-Jun-22	Adjustment	Adjusted 30-06-2022	30-Jun-21	Change	Change %
Assets						
Non-current Assets						
Property, Plant & Equipment	71,411,648	-	71,411,648	79,346,275	(7,934,627)	-10.0%
Right of use assets	14,878,900		14,878,900	16,532,115	(1,653,215)	-10.0%
Investments	2,014,180		2,014,180	2,014,180	-	0.0%
Deferred tax assets	1,514,589		1,514,589	931,386	583,203	62.6%
	89,819,317	**-**	**89,819,317**	**98,823,956**	**(7,934,627)**	**-8.0%**
Current Assets						
Biological assets	20,435,495	2,313,760	22,749,255	19,099,165	3,650,090	19.1%
Inventories	2,942,680		2,942,680	826,330	2,116,350	256.1%
Trade & Other receivables	5,232,283	1,000,000	6,232,283	3,412,042	2,820,241	82.7%
Advance, Deposits & Pre-payments	12,089,350	-	12,089,350	9,410,475	2,678,875	28.5%
Cash & Cash Equivalents	5,514,242	-	5,514,242	2,109,212	3,405,030	161.4%
	46,214,050	**3,313,760**	**49,527,810**	**34,857,224**	**14,670,586**	**42.1%**
	136,033,367	**3,313,760**	**139,347,127**	**133,681,180**	**6,735,959**	**5.0%**
Equity & Liabilities						
Shareholders' Equity						
Share Capital	100,000,000	-	100,000,000	100,000,000	-	-
Retained Earnings	(2,297,882)	2,433,900	136,018	(6,456,193)	6,592,211	-102.1%
	97,702,118	**2,433,900**	**100,136,018**	**93,543,807**	**6,592,211**	**-102.1%**
Non-Current Liabilities						
Long term borrowings	5,090,359	1,486,110	6,576,469	3,309,386	3,267,083	98.7%
Lease liabilities	7,740,749	-	7,740,749	9,656,886	(1,916,137)	-19.8%
Gratuity obligation	8,889,238	900,000	9,789,238	8,453,289	1,335,949	15.8%
	21,720,346	**2,386,110**	**24,106,456**	**21,419,561**	**2,686,895**	**94.7%**
Current Liabilities						
Short Term Borrowings	1,233,498	(1,866,250)	(632,752)	2,835,211	(3,467,963)	-122.3%
Trade and other Payables	6,520,791	-	6,520,791	3,017,267	3,503,524	116.1%
Provisions and other liabilities	5,202,439	360,000	5,562,439	9,833,319	(4,270,880)	-43.4%
Income tax payables	3,654,175	-	3,654,175	3,032,015	622,160	20.5%
	16,610,903	**(1,506,250)**	**15,104,653**	**18,717,812**	**(3,613,159)**	**-29.1%**
	136,033,367	**3,313,760**	**139,347,127**	**133,681,180**	**5,665,947**	**-36.5%**

Appendix 3: Ratio Analysis:

		2021	2020
1	Current Ratio	3.3	1.9
2	Quick Ratio	1.8	0.8
3	Debt to equity ratio	0.24	0.23
4	Asset Turnover Ratio	1.69	1.50
5	TIE Ratio	11.6	3.2
6	Return on Asset	4.7%	1.6%
7	Return on Equity	6.6%	2.3%

Annexure 4: Analysis of the strategic options of setting up biogas plant.

Discount rate derived = 13%.

Year	Source/use of cash	Amount	Discount Rate	Present Value
0	Initial investment	(100,000,000)	1.00	(100,000,000)
1	Sell of biogas	19,775,000	0.88	17,500,000
2	Sell of biogas	19,775,000	0.78	15,486,726
3	Sell of biogas	19,775,000	0.69	13,705,067
4	Sell of biogas	19,775,000	0.61	12,128,378
5	Sell of biogas	19,775,000	0.54	10,733,078
6	Sell of biogas	19,775,000	0.48	9,498,299
7	Sell of biogas	19,775,000	0.43	8,405,574
8	Sell of biogas	19,775,000	0.38	7,438,561
9	Sell of biogas	19,775,000	0.33	6,582,798
10	Sell of biogas	19,775,000	0.29	5,825,485
	Total	**97,750,000**		**7,303,965**

Annexure 5: Analysis of the strategic options of expanding fish production.

Discount rate derived = 13%.

Year	Source/use of cash	Amount	Discount Rate	Present Value
0	Initial investment	(100,000,000)	1.00	(100,000,000)
1	Net revenue from fish	11,000,000	0.88	9,734,513
2	Net revenue from fish	11,000,000	0.78	8,614,614
3	Net revenue from fish	11,000,000	0.69	7,623,552
4	Net revenue from fish	11,000,000	0.61	6,746,506
5	Net revenue from fish	11,000,000	0.54	5,970,359
6	Net revenue from fish	11,000,000	0.48	5,283,504
7	Net revenue from fish	11,000,000	0.43	4,675,667
8	Net revenue from fish	11,000,000	0.38	4,137,758
9	Net revenue from fish	11,000,000	0.33	3,661,733
10	Net revenue from fish	11,000,000	0.29	3,240,472
11	Net revenue from fish	11,000,000	0.26	2,867,674
12	Net revenue from fish	11,000,000	0.23	2,537,765
13	Net revenue from fish	11,000,000	0.20	2,245,810
14	Net revenue from fish	11,000,000	0.18	1,987,442
15	Net revenue from fish	11,000,000	0.16	1,758,798
	Total	**65,000,000**		**(28,913,833)**

Annexure 6: Analysis of the strategic options of setting up biogas plant financing through loan.

Discount rate derived = 12%.

Year	Source/use of cash	Amount	Discount Rate	Present Value
0	Initial investment	(100,000,000)	1.00	(100,000,000)
1	Sell of biogas	19,695,000	0.89	17,584,821
2	Sell of biogas	19,695,000	0.80	15,700,733
3	Sell of biogas	19,695,000	0.71	14,018,512
4	Sell of biogas	19,695,000	0.64	12,516,529
5	Sell of biogas	19,695,000	0.57	11,175,472
6	Sell of biogas	19,695,000	0.51	9,978,100
7	Sell of biogas	19,695,000	0.45	8,909,018
8	Sell of biogas	19,695,000	0.40	7,954,480
9	Sell of biogas	19,695,000	0.36	7,102,214
10	Sell of biogas	19,695,000	0.32	6,341,263
	Total	**96,950,000**		**11,281,143**

Annexure 7: Analysis of the strategic options of setting up biogas plant financed by shares.

Discount rate derived = 13%.

Year	Source/use of cash	Amount	Discount Rate	Present Value
0	Initial investment	(100,000,000)	1.00	(100,000,000)
1	Sell of biogas	19,675,000	0.88	17,411,504
2	Sell of biogas	19,675,000	0.78	15,408,411
3	Sell of biogas	19,675,000	0.69	13,635,762
4	Sell of biogas	19,675,000	0.61	12,067,046
5	Sell of biogas	19,675,000	0.54	10,678,802
6	Sell of biogas	19,675,000	0.48	9,450,267
7	Sell of biogas	19,675,000	0.43	8,363,068
8	Sell of biogas	19,675,000	0.38	7,400,945
9	Sell of biogas	19,675,000	0.33	6,549,509
10	Sell of biogas	19,675,000	0.29	5,796,026
	Total	**96,750,000**		**6,761,340**

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